

Project Overview — Geometric Projection Dynamics

Andrea Sanna · worked example in classical mechanics, differential geometry, and symbolic/numerical verification

This package is a self-contained worked problem in theoretical and computational physics, provided as a representative sample of how I formulate a problem, derive results analytically, and verify them computationally. It is drawn from my independent research programme and has been prepared as a standalone, generic mechanics exercise.

THE PROBLEM

A point moves freely — along a geodesic, i.e. a great circle — on a sphere $S^2(\mathbb{R})$ in three-dimensional space. An observer confined to the equatorial plane sees only the two-dimensional orthogonal projection of that motion. The question is: **what apparent dynamics does the planar observer reconstruct?** A free, force-free motion in 3D, seen in projection, appears to the 2D observer as motion under an apparent force. The exercise is to derive that apparent force rigorously and to characterise it.

WHAT THE WORK DEMONSTRATES

- **Differential geometry & classical mechanics:** parametrising a geodesic on a sphere, projecting it, and deriving the induced equations of motion in the observed plane.
- **The central subtlety — the time map:** the observer's clock must be related to the affine parameter along the geodesic. Three candidate relations are examined; only one is physically admissible (positive, smooth, conserving the observed speed), and it turns out to coincide exactly with the special-relativistic time-dilation factor. Identifying which choice is valid, and why, is the crux of the problem.
- **Closed-form derivation:** apparent velocity, apparent acceleration in matrix form, conservation of observed speed, and an exact invariant $|a| \cdot t^{3/2} = R\omega^2 \cdot \cos i$ that holds around the entire orbit.
- **Emergence of relativistic structure:** a generalised Lorentz factor and special-relativistic time dilation arise purely from the geometry of the projection, with no relativity assumed as input.
- **Dual verification:** every analytical result is confirmed two ways — by an independent symbolic computation (SymPy), with each identity checked to a residual of zero, and by direct numerical integration, with agreement at machine precision ($\sim 10^{-15}$).

CONTENTS OF THIS PACKAGE

- **derivation_note.pdf** — the full written derivation: setup, the three time maps, the closed-form dynamics, the invariant, and the time-dilation result, with proofs.
- **turing_sphere_projection.py** — numerical implementation. Builds the geodesic, compares the three time maps, computes the closed-form velocity and acceleration, and cross-checks every result against a numerical derivative. Generates the figures.
- **turing_symbolic_projection_derivation.py** — symbolic derivation in SymPy of the projected observables, with each algebraic identity verified to exactly zero.

METHOD NOTE

My research practice combines analytical derivation, symbolic verification, and numerical cross-checking, with each result required to survive at least two independent methods before it is accepted. The two scripts here reflect

that: the closed forms in the written note are reproduced independently in symbolic and numerical form, and only results that agree to machine precision are reported. This is the same evaluative discipline I apply when assessing the correctness of a chain of physical reasoning.

A formal record of related published work is available on Zenodo ([DOI 10.5281/zenodo.19469337](https://doi.org/10.5281/zenodo.19469337)).